Hybrid Feature Selection and Calibration-Aware Ensemble Learning for Explainable IoT Intrusion Detection and Classification

***Abstract*—Reliable intrusion detection in Internet of Things (IoT) networks requires not only high classification accuracy but also interpretable predictions and trustworthy confidence estimates. This paper proposes Hybrid Ensemble Feature Selection (HEFS), which combines mutual information, XGBoost gain, recursive feature elimination, and SHAP-based feature importance through average-rank aggregation to select 20 interpretable network traffic features. Using the ToN-IoT dataset, HEFS is evaluated against a 23-component Principal Component Analysis (PCA) baseline across seven classifiers for binary and ten-class intrusion classification. To prevent data leakage into the hold-out test set, SMOTE and feature scaling are restricted to the training partition. HEFS improves ten-class classification accuracy across all seven classifiers while maintaining comparable performance in binary classification. Random Forest with HEFS achieves the highest hold-out accuracy of 99.91% for binary classification and 98.93% for ten-class classification. SHAP analysis identifies destination ports, protocol, packet and byte counts, and connection-state attributes as influential features. Temperature scaling reduces the Random Forest's expected calibration error from 0.0096 to 0.0021. Furthermore, referring the 5% least-confident evaluation flows for manual review increases accuracy on the remaining flows from 98.88% to 99.88%. However, the rare man-in-the-middle (MITM) class remains challenging, with a precision of 0.60. These findings demonstrate that HEFS provides an interpretable alternative to PCA, while probability calibration and selective prediction support more reliable, human-in-the-loop IoT intrusion detection.**

***Index Terms*—intrusion detection, IoT security, ToN-IoT, explainable AI, SHAP, feature selection, probability calibration**

I. Introduction

The Internet of Things (IoT) is made up of billions of resource constrained sensors and actuators, many with default credentials and unpatched firmware, that are appealing and easy targets. A main line of defence is network intrusion detection systems (IDS), and recent learning-based detectors, like the tree ensembles, gradient boosting, and tabular deep networks, have reached an accuracy of over 95% on public IoT benchmarks like ToN-IoT [**?**], [1]. But just accuracy isn't enough for a detector to be deployable. If the analyst receives an "Attack/DDoS" alert, the analyst needs to know what attributes of the traffic were used to generate the alert in order to confirm the alert, prioritize a response and eliminate false alerts, and a black-box model provides no such information.

This isn't as easy as it sounds for three reasons. The common preprocessing step PCA transforms each input into a linear combination of a large number of flow attributes, hence later explanations use the word component instead of network feature names. Second, IoT traffic is highly imbalanced – ToN-IoT is 48 times less than the normal traffic, oversampling before the train/test split causes a leakage of synthetic neighbours of test points into the training set, which leads to an inflated accuracy. Third, alert systems are enabled by confidence thresholds that assumes trustworthy predicted probabilities, something that trees, deep networks and SMOTE-trained models do not always possess [3].

Previous research largely resolves this dichotomy by having a feature-extraction pipeline that is complex and opaque for accuracy, or very small for interpretability with reduced performance on the rare classes (Section II). In this paper, there are four contributions.

1) **HEFS.** A hybrid feature-selection scheme that fuses filter, embedded, wrapper and SHAP rankings by average rank into 20 *named* flow features, so the same SHAP statistic drives both feature selection and the final explanation.
2) **Like-for-like benchmark.** A leakage-safe comparison of PCA and HEFS over seven classifiers, on a binary and a ten-class task, with one split and identical cross-validation folds.
3) **Statistical evidence.** Corrected resampled t-tests, Friedman–Nemenyi, McNemar's test and a paired bootstrap with Holm correction, rather than raw accuracy gaps.
4) **Trust analysis.** Global and local SHAP explanations of the best model, plus calibration (temperature scaling, isotonic regression) and a selective-prediction operating point for human-in-the-loop deployment.

II. Related Work

A summary of ToN-IoT and related works is given in Table I. The feature-selection approaches include filter [4], embedded [5], wrapper and metaheuristic methods, while the accuracy-oriented approaches involve the use of attention or CNN-LSTM extractors [6]–[8], quantum inspired weighted ensembles [9] and spiking networks [10]. Typically, explainability is added on one step: [4] explains a universal set of 6 features using SHAP, while [11] explains an unsupervised detector using LIME. There are two failure modes reported that motivate our design. The aggressiveness of the reduction to 5 features causes a drop in the precision of the MITM model ([12]) and it is possible that a weak MITM model (78% F1) is not diagnosed by any model in the pipeline [13]. Models

TABLE I
RELATED TON-IOT INTRUSION-DETECTION STUDIES

Study	Approach, reported result	Gap
Alani & Miri [4]	6-feature filter + SHAP, 96.62% (DT)	One classifier only
Disha & Waheed [5]	Embedded GIWRF, 10 features	Binary only; no XAI
Maseno et al. [7]	CNN-LSTM reduction + RF, 98%	Opaque reduction
Shaikhanova et al. [13]	RF/LightGBM/XGBoost benchmark, 99%	MITM F1 78%; no XAI
Dharini et al. [12]	Bias-aware 5-feature selection + DT, 97.20% (10-class)	Lower MITM precision; no SHAP check
Hossain et al. [14]	RF, 99.79% in-dataset	<40% cross-dataset
This work	HEFS (20 named features) + SHAP + calibration, 7 models, 2 tasks	Single dataset

that perform very well in-dataset can collapse across datasets [14].

None of the studies reviewed (i) applies one SHAP rationale for feature selection and final explanation or (ii) compares PCA with a fully named hybrid feature selection scheme using the same data, split and classifiers, or (iii) reports the calibration of the detector. This study address (i)–(iii). Cross data set generalization is open.

III. METHODOLOGY

A. Dataset

The ToN-IoT network dataset (`train_test_network.csv`) [**?**], [1] consists of 211,043 flow records (161,043 attack, 50,000 normal), 44 attributes including connection, DNS, SSL/TLS, HTTP and Zeek anomaly fields, a binary label and an attack-type column. There are a total of ten classes in the type column: normal traffic and nine attacks. The 8 attack types include 20,000 flows each, whereas MITM only has 1,043, giving it 48 times fewer than the normal traffic, and 19 times fewer than the other attack types. Both the source and destination IP addresses are discarded as a classifier might otherwise have to remember attacker hosts, while both the label columns are also dropped, leaving 40 candidate features (16 numeric and 24 categorical).

B. Leakage-Safe Preprocessing

The placeholder "-" becomes NaN; numeric NaNs are filled with 0 and categorical NaNs with a "missing" level, so the absence of, for example, HTTP activity remains a signal. Categorical columns are label-encoded. The data are split once, stratified 80:20 (seed 42), into 168,834 training and 42,209 test flows. SMOTE [15] (k=5) is applied to the training partition only: the binary training set becomes 128,834 flows per class, and every class of the ten-class set becomes 40,000 flows (MITM: 834 $\rightarrow$ 40,000). The test set is never resampled. Features are standardised with a scaler fitted on training data.

C. Dimensionality Reduction

PCA baseline. We select the minimum number of components to explain 95% variance, 23 components for both tasks (a 42.5% reduction). No component has any direct network meaning and is a linear combination of the original features.

HEFS (proposed). Four rankings are computed on the standardised training features (Fig. 2): (1) *filter*, mutual information; (2) *embedded*, total gain of a fitted XGBoost model; (3) *wrapper*, recursive feature elimination (RFE) with an XGBoost estimator; and (4) *explainable*, mean absolute SHAP value from a TreeExplainer on the same XGBoost model. Each method has its own bias (mutual information ignores interactions, gain favours high-cardinality features, RFE depends on the estimator), so the ranks are fused by simple average:

$$\mathrm{AvgRank}(f) = \frac{1}{4} \sum_{m \in \{\mathrm{MI,XGB,RFE,SHAP}\}} \mathrm{rank}_m(f). \quad (1)$$

The 20 features with the lowest average rank are kept, separately for each task. Every retained feature keeps its original name, so explanations read as statements about real traffic attributes. For the binary task the top five are `dst_port`, `src_port`, `proto`, `src_pkts` and `conn_state`.

D. Classifiers

The classifiers trained on both representations and both tasks are: Random Forest (RF; 50 trees, min. leaf 2) [16]; k-nearest neighbours (k=5) [**?**]; XGBoost (100 trees, depth 6, learning rate 0.1) [18]; CatBoost (200 iterations, depth 6, learning rate 0.1) [19]; TabTransformer [20]; an autoencoder-based classifier; and a temporal convolutional network (TCN) [21]. The three deep models are trained for 30 epochs with a validation split of 10%.

E. Evaluation Protocol

Accuracy, per-class precision, recall and F1, macro and weighted F1 (and ROC-AUC, PR-AUC for binary task) are reported on the default test set that has not been used for training. For the deep models, five-fold stratified cross-validation (CV) is conducted on the representation of the HEFS (as opposed to 30 epochs in this case). The significance is estimated by a corrected resampled t-test [22], a Friedman test with post-hoc Nemenyi [23] and a McNemar's test and a paired bootstrap (2,000 resamples) applied to the test set, all of which are followed by a Holm correction for the six comparisons with RF. Expected and maximum calibration error (ECE, MCE; 15 bins), Brier score and negative log-likelihood are used for calibration. The 30% of the test flows included in the stratified temperature scaling and isotonic regression are the ones used to fit the temperature scales and the remaining 70% (n=29,547) are the ones used for a report.

F. Explainability

TreeExplainer [24] gives exact SHAP values for tree models [25]; KernelExplainer is the model-agnostic fallback. Each

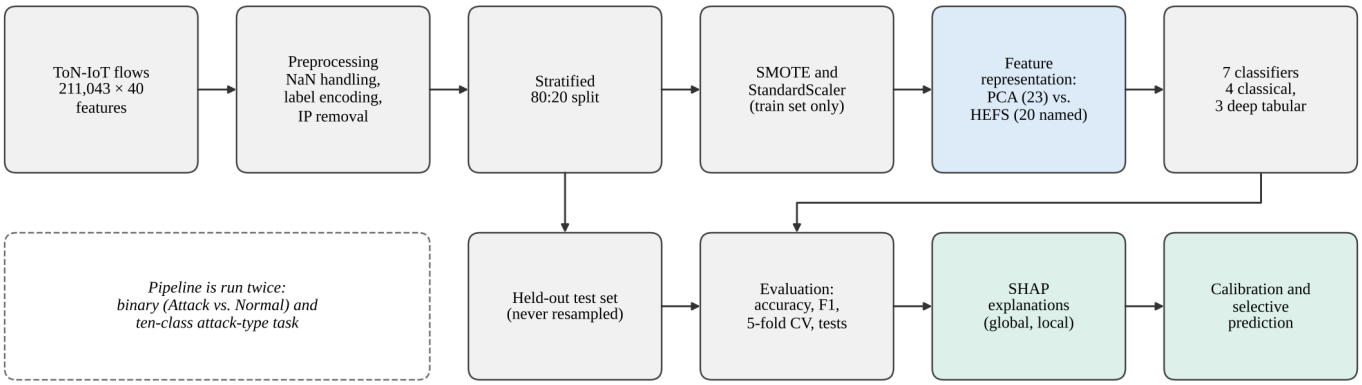


Fig. 1. Overview of the leakage-safe pipeline. SMOTE and scaling touch the training partition only; the held-out test set is never resampled. HEFS (blue) is the proposed representation; SHAP and calibration (green) analyse the best model.

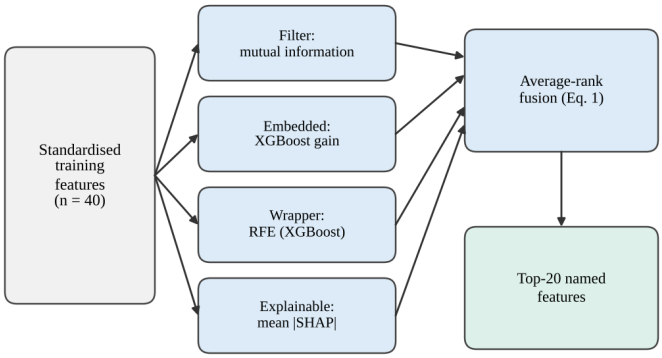


Fig. 2. HEFS: four ranking branches are fused by average rank (Eq. 1) into a 20-feature named set.

explanation is additive, $f(x) = \varphi_0 + \sum_i \varphi_i$, and global importance is

$$I_j = \frac{1}{N} \sum_{k=1}^{N} \left| \varphi_j^{(k)} \right|, \tag{2}$$

the same statistic as the SHAP branch of HEFS. Local explanations use waterfall plots for individual flows.

IV. Results and Discussion

A. Accuracy: PCA vs. HEFS

The hold-out accuracy of all 28 models is provided in Table II and the 5-fold CV accuracy for the HEFS models for the 10-class task is provided. In terms of the binary classification task, RF on HEFS achieves 99.91% accuracy (ROC-AUC 0.9999, PR-AUC 1.0000; 15 attacks misclassified, 24 normal flows misclassified out of 42,209).For the ten-class classification task, the accuracy achieved by RF on HEFS is 98.93%, outperforming the other methods. For the binary task, the representation is hardly relevant, with HEFS being able to aid by 0.18 points for the tree ensembles, as well as the TCN, and only slightly worse than PCA for kNN, TabTransformer, and the autoencoder. On the harder ten-class task HEFS wins

TABLE II
Hold-out Accuracy by Representation and 5-Fold CV (Ten-Class, HEFS)

	Binary		Ten-class		10-class CV
Model	PCA	HEFS	PCA	HEFS	(HEFS)
Random Forest	0.9980	**0.9991**	0.9776	**0.9893**	0.9921 ± .0003
XGBoost	0.9969	0.9987	0.9621	0.9876	0.9898 ± .0003
CatBoost	0.9968	0.9984	0.9368	0.9845	0.9856 ± .0005
KNN	0.9975	0.9975	0.9724	0.9736	0.9752 ± .0003
Autoencoder	0.9951	0.9948	0.9308	0.9617	0.9323 ± .0077
TabTransformer	0.9954	0.9952	0.9368	0.9609	0.9322 ± .0044
TCN	0.9932	0.9950	0.8868	0.9357	0.8771 ± .0100

for every model, by 0.12 points for kNN up to 4.77 and 4.89 points for CatBoost and the TCN. We interpret this as evidence that the PCA leading directions don't capture the differences between attack types, but named flow attributes do. CV ranks the models in the same order as the hold-out set, and the deep models vary considerably from fold to fold (at most up to 0.0100 for TCN, compared to up to 0.0005 for the tree models).

B. Per-Class Behaviour

The per-class report of RF and confusion matrix are shown in Table III and Fig. 3, respectively. The best performance seen in the 70% of the classes is F1 $\geq$ 0.99 and for injection and XSS is 0.98 with injection pulldown (precision) having a value of 0.60 and XSS pulldown (recall) 0.91 and macro F1 0.72. There are 128 flows predicted as other classes (48 DoS, 32 password, 19 normal, 17 scanning, 8 injection, 4 DDoS) of the 318 flows. Since there were 834 actual flows and 48-fold oversampling for MITM, it is likely that the model overpredicts MITM, which is an aspect we did not test. This pipeline is the least strong in the rare class, not the overall accuracy.

C. Statistical Significance

RF has an edge over each of its competitors at both levels of evaluation. The corrected re-sampled t-test for the 5 folds

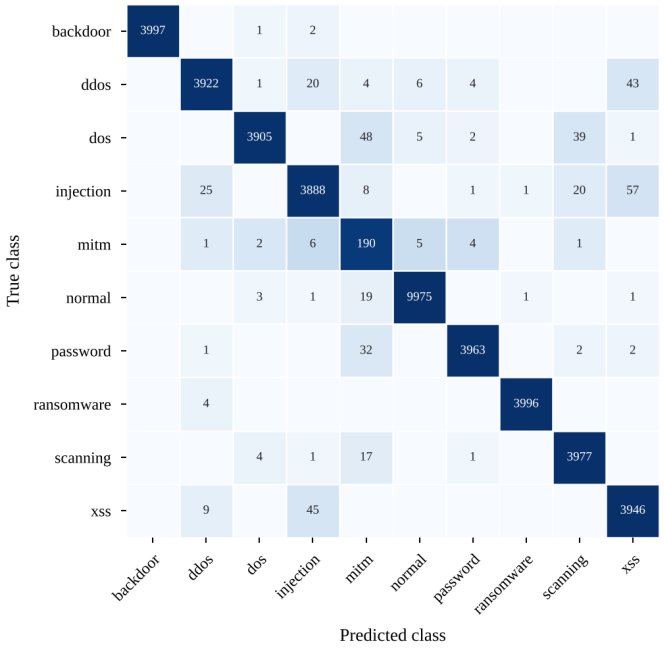


Fig. 3. RF (HEFS) confusion matrix, ten-class task, hold-out set (n=42,209). Cell values are counts; shading is row-normalised recall.

TABLE III
RF (HEFS) PER-CLASS RESULTS, TEN-CLASS HOLD-OUT SET

Class	Precision	Recall	F1	Support
backdoor	1.00	1.00	1.00	4,000
DDoS	0.99	0.98	0.99	4,000
DoS	1.00	0.98	0.99	4,000
injection	0.98	0.97	0.98	4,000
MITM	0.60	0.91	0.72	209
normal	1.00	1.00	1.00	10,000
password	1.00	0.99	0.99	4,000
ransomware	1.00	1.00	1.00	4,000
scanning	0.98	0.99	0.99	4,000
XSS	0.97	0.99	0.98	4,000
macro avg	0.95	0.98	0.96	42,209
weighted avg	0.99	0.99	0.99	42,209

of the CV shows that all 6 comparisons have $p \leq 0.0032$ (smallest margin: XGBoost, 0.23 points, $p = 0.0017$), and the Friedman test yields a $p = 0.00005$ which rejects equality. Limited power of the Nemenyi test means that the top 4 models cannot be distinguished with just five folds (critical difference 4.03), this is not proof of equivalence. For all six comparisons, both McNemar's test and the paired bootstrap agree that the two distributions are not equal: Holm adjusted $p < 0.001$; in every Bootstrap interval, the value of K does not include 0. But there's not much difference between Agility and XGBoost in practice; on the hold-out set, Agility was correct on 125 flows, while XGBoost was correct on 52.

D. Explainability

Fig. 4 shows global SHAP importance for RF (HEFS). For attack vs. normal the dominant features are destination port, protocol, source packet count and a DNS rejection flag. For the ten-class task, byte volume (`src_ip_bytes`), ports, connection state, and duration rise to the top while DNS features fall, matching the intuition that separating attack types depends on traffic volume and connection behavior, whereas separating attack from normal needs only a few strong signals. Waterfall plots for individual flows decompose each prediction into the contributions of named features around the model's base rate. A caution applies: port-based features can encode testbed artifacts rather than attack behavior, and our experiments cannot rule this out (Section IV-G).

E. Calibration and Selective Prediction

XGBoost is well calibrated (ECE 0.0023) out of the box. Even with the highest accuracy, RF is only middling (ECE 0.0095) with a slight under-confidence (-0.0091). For six of the seven models, ECE decreases with temperature scaling (see Fig. 5a); for RF, temperature is fitted to 0.752, and ECE decreases from 0.0096 to 0.0021. Isotonic regression is even better for RF (ECE 0.0018, MCE 0.1425), but still temperature scaling will cause MCE to increase (0.253 to 0.738) as it optimises the average gap, and isn't as good as temperature scaling since it does not ensure the same accuracy. Per class: 8 out of 10 classes are improved by temperature scaling, but not by MITM (0.0019 to 0.0021) or injection as there are only 209 test flows.

The operational payoff is selective prediction (Fig. 5b). With the 5% least-confident flows (1,476 out of 29,547) manually scaled by the temperature-scaled RF and then passed to an analyst, the rest of the flows will be scaled with 99.88% accuracy, compared to 98.88% when the entire flow-set was passed to the RF.

F. Discussion

Three observations follow. First, interpretability did not incur any loss of accuracy: on the binary task, HEFS and PCA are essentially equal, and on the ten-class task, the named representation is better for all seven models. Second, the most accurate model was not the best calibrated; accuracy should not be the only criteria used to rank the models if there are confidence thresholds to set.Second, the most accurate model was not the best calibrated, so an accuracy ranking would lead a team to believe that this model would be more accurate if there were confidence thresholds to set. Thirdly, the weakness of the pipeline exists in a very small class which the accuracy of 98.93% could mask, and which was not corrected with the use of SMOTE or temperature scaling.

G. Limitations

No cross-dataset test: All results are from one dataset and one split thereof, as shown in [14] how large that gap could be. (iii) CV was performed on the SMOTE resampled training set and the McNemar and bootstrap results were performed using the clean hold-out set and are not affected; CV accuracy is optimistically biased by using the SMOTE resampled training set prior to the fitting of HEFS. The explanations are dominated

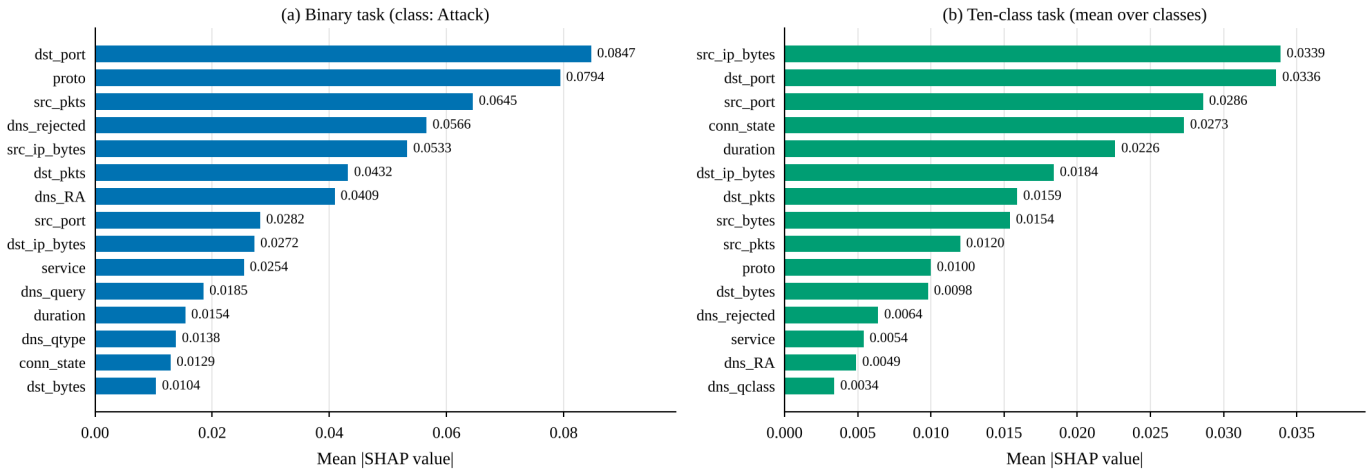


Fig. 4. Global SHAP importance (mean |SHAP|) for RF on HEFS features: (a) binary task, positive class Attack; (b) ten-class task, averaged over classes.

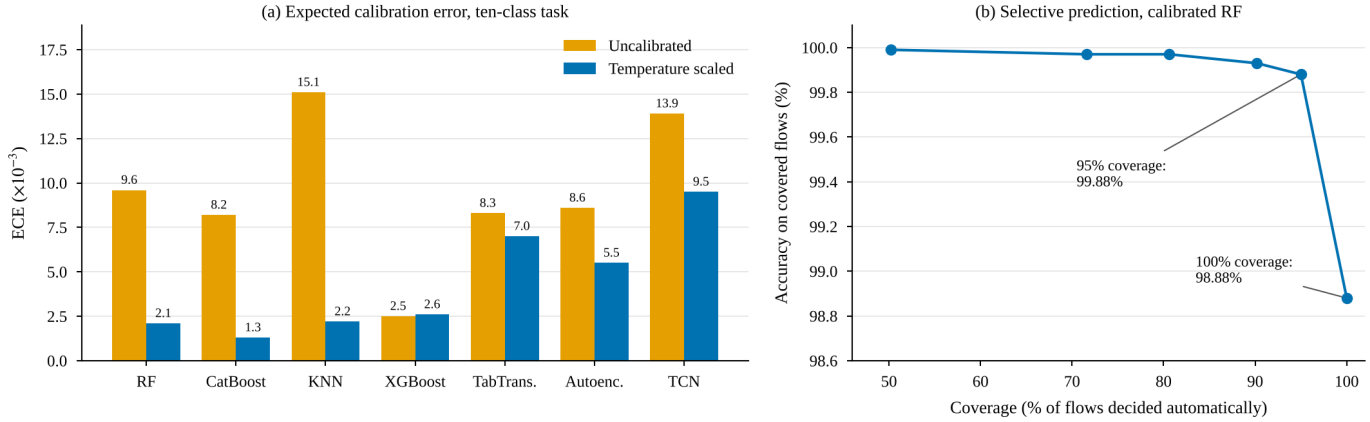


Fig. 5. (a) ECE before and after temperature scaling, ten-class task, evaluation split (n=29,547). (b) Accuracy on covered flows when the temperature-scaled RF abstains on its least-confident flows.

by port features (iii) and may be indicative of the testbed. There was no resolution for (iv) MITM precision (0.60). (v) Related work has different subsets, splits and class definitions, and reported accuracy ranges from 97.20% ten-class [12] to 98% [7] so we cannot claim a direct ranking against them. Only the best model was used to compute SHAP in (vi).

V. Conclusion and Future Work

This study proposed HEFS, a four-branch average-rank feature-selection scheme which provides 20 named flow features, and compared it to PCA on seven classifiers in the context of binary and ten-class intrusion detection under leakage-safe protocol for ToN-IoT. Random Forest on HEFS reached 99.91% and 98.93% hold-out accuracy, and was significantly better than all six alternatives, SHAP tied decisions to security-relevant attributes, and calibration made human-in-the-loop operating point possible (99.88% accuracy on 95% of flows). This was not a case of accuracy vs interpretation, as the easy task is the more accurate. Future research will focus on cost sensitive learning, MITM, focal loss or oversampling using ADASYN, extension of SHAP to all models, testing HEFS on CICIoT2023, Edge-IIoTset and NSL-KDD, measurement of inference latency & throughput in deployed service, and evaluation of stacking of strongest models.

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